



SAFETY DATA SHEET

RETAIL FREEZER/COOLER CLEANER

SECTION 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : RETAIL FREEZER/COOLER CLEANER

Other means of identification : not applicable

Recommended use : Cleaning product

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : Product is sold ready to use.

Company : Kay Chemical Company
8300 Capital Drive
Greensboro, North Carolina USA 27409
1-800-529-5458

Emergency telephone : 1-800-328-0026 (US/Canada), 1-651-222-5352 (outside US)

Issuing date : 10.06.2014

SECTION 2. HAZARDS IDENTIFICATION

Physical and chemical hazards : Flammable liquid and vapor.

Potential Health Effects

Eyes : Causes eye irritation.

Skin : Causes mild skin irritation.

Ingestion : Health injuries are not known or expected under normal use.

Inhalation : Health injuries are not known or expected under normal use.

Chronic Exposure : Health injuries are not known or expected under normal use.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Pure substance/mixture : Mixture

Chemical Name	CAS-No.	Concentration (%)
glycerin	56-81-5	30 - 60
propan-2-ol	67-63-0	5 - 10
ethanolamine	141-43-5	1 - 5
poly(oxy-1,2-ethanediyl), .alpha.-undecyl-.omega.-hydroxy-	34398-01-1	1 - 5

SECTION 4. FIRST AID MEASURES

In case of eye contact : Rinse with plenty of water.

In case of skin contact : Rinse with plenty of water.

If swallowed : Rinse mouth. Get medical attention if symptoms occur.

If inhaled : Get medical attention if symptoms occur.

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Protection of first-aiders : No special precautions are necessary for first aid responders.

Notes to physician : Treat symptomatically.

See toxicological information (Section 11)

SECTION 5. FIRE-FIGHTING MEASURES

- Suitable extinguishing media : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
- Unsuitable extinguishing media : High volume water jet
- Specific hazards during fire fighting : Fire Hazard
Keep away from heat and sources of ignition.
Flash back possible over considerable distance.
Beware of vapors accumulating to form explosive concentrations.
Vapors can accumulate in low areas.
- Hazardous combustion products : Decomposition products may include the following materials:
Carbon oxides
nitrogen oxides (NOx)
Sulfur oxides
Oxides of phosphorus
- Special protective equipment for fire-fighters : Use personal protective equipment.
- Specific extinguishing methods : Use water spray to cool unopened containers. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

SECTION 6. ACCIDENTAL RELEASE MEASURES

- Personal precautions, protective equipment and emergency procedures : Remove all sources of ignition. Refer to protective measures listed in sections 7 and 8.
- Environmental precautions : Do not allow contact with soil, surface or ground water.
- Methods and materials for containment and cleaning up : Eliminate all ignition sources if safe to do so. Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

SECTION 7. HANDLING AND STORAGE

- Advice on safe handling : Use only with adequate ventilation. Keep away from fire, sparks and heated surfaces. Take necessary action to avoid static electricity discharge (which might cause ignition of organic vapors). Wash hands thoroughly after handling.
- Conditions for safe storage : Keep away from heat and sources of ignition. Keep in a cool, well-ventilated place. Keep away from oxidizing agents. Keep out of reach

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of children. Keep container tightly closed. Store in suitable labeled containers.

Storage temperature : 0 °C to 35 °C

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Ingredients	CAS-No.	Form of exposure	Permissible concentration	Basis
glycerin	56-81-5	CMP	10 mg/m3	AR OEL
glycerin	56-81-5	TWA	10 mg/m3	PE OEL
glycerin	56-81-5	CAP	10 mg/m3	VE OEL
glycerin	56-81-5	CPT (Total dust)	10 mg/m3	PA OEL
glycerin	56-81-5	LMPE-PPT	10 mg/m3	MX OEL
glycerin	56-81-5	TWA	10 mg/m3	ACGIH
propan-2-ol	67-63-0	CMP	400 ppm	AR OEL
		CMP - CPT	500 ppm	AR OEL
propan-2-ol	67-63-0	LT	310 ppm 765 mg/m3	BR OEL
propan-2-ol	67-63-0	LPP	320 ppm 786 mg/m3	CL OEL
		LPT	500 ppm 1,230 mg/m3	CL OEL
propan-2-ol	67-63-0	TWA	200 ppm 491 mg/m3	PE OEL
		STEL	400 ppm 983 mg/m3	PE OEL
propan-2-ol	67-63-0	CAP	400 ppm	VE OEL
		LEB	500 ppm	VE OEL
propan-2-ol	67-63-0	CPT	400 ppm 960 mg/m3	PA OEL
		CCT	500 ppm 1,225 mg/m3	PA OEL
propan-2-ol	67-63-0	LMPE-PPT	400 ppm 980 mg/m3	MX OEL
		LMPE-CT	500 ppm 1,225 mg/m3	MX OEL
propan-2-ol	67-63-0	TWA	200 ppm	ACGIH
		STEL	400 ppm	ACGIH
		TWA	400 ppm 980 mg/m3	NIOSH REL
		STEL	500 ppm 1,225 mg/m3	NIOSH REL
		TWA	400 ppm 980 mg/m3	OSHA Z1
ethanolamine	141-43-5	CMP	3 ppm	AR OEL
		CMP - CPT	6 ppm	AR OEL
ethanolamine	141-43-5	TWA	3 ppm 7.5 mg/m3	PE OEL
		STEL	6 ppm 15 mg/m3	PE OEL
ethanolamine	141-43-5	CAP	3 ppm	VE OEL
		LEB	6 ppm	VE OEL

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ethanolamine	141-43-5	CCT	6 ppm 15 mg/m3	PA OEL
		CPT	3 ppm 6 mg/m3	PA OEL
ethanolamine	141-43-5	LMPE-PPT	3 ppm 8 mg/m3	MX OEL
		LMPE-CT	6 ppm 15 mg/m3	MX OEL
ethanolamine	141-43-5	TWA	3 ppm	ACGIH
		STEL	6 ppm	ACGIH
		TWA	3 ppm 8 mg/m3	NIOSH REL
		STEL	6 ppm 15 mg/m3	NIOSH REL
		TWA	3 ppm 6 mg/m3	OSHA Z1

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : No special protective equipment required.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Wash face, hands and any exposed skin thoroughly after handling.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	: liquid
Color	: cloudy, light green
Odor	: alcoholic
pH	: 10.6 - 11.8, 100 %
Flash point	: 41 °C closed cup, Sustains combustion
Odor Threshold	: no data available
Melting point/freezing point	: no data available
Initial boiling point and boiling range	: no data available
Evaporation rate	: no data available
Flammability (solid, gas)	: no data available
Upper explosion limit	: no data available
Lower explosion limit	: no data available
Vapor pressure	: no data available
Relative vapor density	: no data available

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Relative density	: 1.065 - 1.085
Water solubility	: soluble
Solubility in other solvents	: no data available
Partition coefficient: n-octanol/water	: no data available
Autoignition temperature	: no data available
Thermal decomposition	: no data available
Viscosity, kinematic	: no data available
Explosive properties	: no data available
Oxidizing properties	: no data available
Molecular weight	: no data available
VOC	: no data available

SECTION 10. STABILITY AND REACTIVITY

Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use.
Conditions to avoid	: Heat, flames and sparks.
Incompatible materials	: Acids
Hazardous decomposition products	: Decomposition products may include the following materials: Carbon oxides nitrogen oxides (NOx) Sulfur oxides Oxides of phosphorus

SECTION 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Potential Health Effects

Eyes	: Causes eye irritation.
Skin	: Causes mild skin irritation.
Ingestion	: Health injuries are not known or expected under normal use.
Inhalation	: Health injuries are not known or expected under normal use.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Experience with human exposure

Eye contact	: Redness, Irritation
Skin contact	: slight irritation
Ingestion	: No symptoms known or expected.

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Inhalation : No symptoms known or expected.

Toxicity

Acute oral toxicity : no data available

Acute inhalation toxicity : no data available

Acute dermal toxicity : no data available

Skin corrosion/irritation : no data available

Serious eye damage/eye irritation : Mild eye irritation

Respiratory or skin sensitization : no data available

Carcinogenicity : no data available

Reproductive effects : no data available

Germ cell mutagenicity : no data available

Teratogenicity : no data available

STOT-single exposure : no data available

STOT-repeated exposure : no data available

Aspiration toxicity : no data available

Ingredients

Acute oral toxicity : glycerin
LD50 rat: 18,300 mg/kg

propan-2-ol
LD50 rat: 4,710 mg/kg

ethanolamine
LD50 rat: 1,089 mg/kg

poly(oxy-1,2-ethanediyl), .alpha.-undecyl-.omega.-hydroxy-
LD50 rat: 1,700 mg/kg

Ingredients

Acute inhalation toxicity : propan-2-ol
4 h LC50 rat: 30 mg/l

Ingredients

Acute dermal toxicity : glycerin
LD50 rabbit: 23,000 mg/kg

propan-2-ol
LD50 rabbit: 12,870 mg/kg

ethanolamine
LD50 rabbit: 1,025 mg/kg

poly(oxy-1,2-ethanediyl), .alpha.-undecyl-.omega.-hydroxy-
LD50 rabbit: > 2,000 mg/kg

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SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity

Environmental Effects : This product has no known ecotoxicological effects.

Product

Toxicity to fish : no data available

Toxicity to daphnia and other aquatic invertebrates : no data available

Toxicity to algae : no data available

Ingredients

Toxicity to fish : glycerin
96 h LC50 Fish : 855 mg/l

propan-2-ol
96 h LC50 Fish : 9,640 mg/l

poly(oxy-1,2-ethanediyl), .alpha.-undecyl-.omega.-hydroxy-
96 h LC50 Fish : 3.2 mg/l

Ingredients

Toxicity to daphnia and other aquatic invertebrates : ethanolamine
48 h EC50 Daphnia : 65 mg/l

Persistence and degradability

no data available

Bioaccumulative potential

no data available

Mobility in soil

no data available

Other adverse effects

no data available

SECTION 13. DISPOSAL CONSIDERATIONS

Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers.

SECTION 14. TRANSPORT INFORMATION

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Sea transport (IMDG/IMO)

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UN number	:	1993
Description of the goods	:	FLAMMABLE LIQUID, N.O.S. (Isopropanol)
Class	:	3
Packing group	:	III
Marine pollutant	:	no

SECTION 15. REGULATORY INFORMATION

The ingredients of this product are reported in the following inventories:

1907/2006 (EU) :
not determined

Switzerland. New notified substances and declared preparations :
The mixture contains substances listed on the Swiss Inventory not determined

United States TSCA Inventory :
On TSCA Inventory

Canadian Domestic Substances List (DSL) :
All components of this product are on the Canadian DSL.

Australia Inventory of Chemical Substances (AICS) :
On the inventory, or in compliance with the inventory

New Zealand. Inventory of Chemical Substances :
On the inventory, or in compliance with the inventory

Japan. ENCS - Existing and New Chemical Substances Inventory :
not determined

Japan. ISHL - Inventory of Chemical Substances (METI) :
not determined

Korea. Korean Existing Chemicals Inventory (KECI) :
On the inventory, or in compliance with the inventory

Philippines Inventory of Chemicals and Chemical Substances (PICCS) :
On the inventory, or in compliance with the inventory

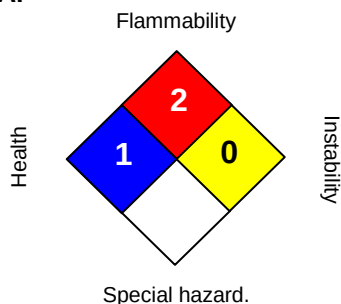
China. Inventory of Existing Chemical Substances in China (IECSC) :
On the inventory, or in compliance with the inventory

SECTION 16. OTHER INFORMATION

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NFPA:



HMIS III:

HEALTH	1
FLAMMABILITY	2
PHYSICAL HAZARD	0

0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Issuing date : 10.06.2014
Version : 1.0
Prepared by : Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the MSDS.

The information provided in this Material Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.